NCLEX Final Review

Executive Summary

This comprehensive review collates the targeted clinical gaps identified in your testing data, creating a single, printable document. Priority areas include Adult Endocrine, Adult Infectious Disease, Pediatric Hematology/Oncology, Cardiovascular, and Musculoskeletal systems. Each section utilizes the ADPIE framework to structure clinical judgment and ensures exhaustive coverage of essential pathologies, diagnostics, and pharmacological interventions without abbreviation.

🟪 Priority Topic 1: Adult Health - Endocrine System

🔴 Pathology Focus: Diabetic Ketoacidosis (DKA) vs. Hyperosmolar Hyperglycemic State (HHS)

DKA primarily affects Type 1 Diabetics, characterized by absolute insulin deficiency, rapid onset, and ketosis. HHS primarily affects Type 2 Diabetics, characterized by profound dehydration, gradual onset, and no ketosis.

🟢 Diagnostic Labs & Monitoring

- **DKA:**
  - **Blood Glucose:** 250 - 800 mg/dL.
  - **Arterial Blood Gas (ABG):** Metabolic acidosis (pH < 7.30, HCO3 < 18 mEq/L).
  - **Urinalysis:** Positive for ketones and glucose.
  - **Serum Potassium:** Initially elevated or normal, then drops rapidly with treatment.
- **HHS:**
  - **Blood Glucose:** > 600 mg/dL (often > 1000 mg/dL).
  - **Serum Osmolality:** > 320 mOsm/kg.
  - **Urinalysis:** Negative for ketones.

🔵 Pharmacological Interventions

- **Intravenous Fluids (IVF):** Initiate with 0.9% Normal Saline (Isotonic) to restore volume. Switch to 5% Dextrose with 0.45% Normal Saline (D5W 1/2 NS) when glucose reaches ~250 mg/dL to prevent hypoglycemia.
- **Regular Insulin (IV):** The only insulin that can be administered intravenously. Administer as a continuous infusion.
- **Potassium Chloride (KCl):** Add to IV fluids once urine output is adequate, as insulin drives potassium back into the cells, causing severe hypokalemia.

🟦 ADPIE Case Study: The DKA Patient

- **Assessment:** Kussmaul respirations, fruity breath odor, lethargy, poor skin turgor, tachycardia.
- **Diagnosis:** Deficient fluid volume related to osmotic diuresis; Risk for electrolyte imbalance.
- **Planning:** Restore fluid volume, correct acidosis, normalize blood glucose levels slowly.
- **Implementation:** Establish two large-bore IVs. Administer bolus of 0.9% Normal Saline. Prime IV tubing with 🔵 **Regular Insulin** before initiating drip. Monitor 🟢 **Blood Glucose** hourly. Monitor telemetry for peaked T-waves (hyperkalemia) or prominent U-waves (hypokalemia).
- **Evaluation:** Resolution of acidosis (HCO3 > 18 mEq/L, pH > 7.30), patient alert and oriented, blood glucose stable between 150-200 mg/dL.

Priority Topic 2: Adult Health - Infectious Disease

Pathology Focus: Sepsis & Septic Shock

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection. Septic shock is a subset of sepsis with profound circulatory, cellular, and metabolic abnormalities.

Diagnostic Labs & Monitoring

- **Lactic Acid (Lactate):** > 2 mmol/L indicates tissue hypoperfusion.
- **Blood Cultures:** Obtain *before* administering antibiotics.
- **White Blood Cell (WBC) Count:** > 12,000/mm³ or < 4,000/mm³, or > 10% immature bands.
- **Procalcitonin:** Elevated levels strongly indicate bacterial infection.

Pharmacological Interventions

- **Broad-Spectrum Antibiotics:** e.g., **Piperacillin/Tazobactam**, **Vancomycin.** Administer within the first hour of recognition.
- **Vasopressors:** **Norepinephrine** is the first-line agent if hypotension persists after adequate fluid resuscitation.
- **Corticosteroids:** **Hydrocortisone** may be considered for patients with refractory shock who do not respond to fluids and vasopressors.

ADPIE Case Study: The Septic Patient

- **Assessment:** Temp > 38.3°C or < 36.0°C, Heart Rate > 90 bpm, Respiratory Rate > 20/min, altered mental status, MAP < 65 mmHg.
- **Diagnosis:** Impaired tissue perfusion; Infection.
- **Planning:** Administer 1-hour bundle interventions. Restore MAP to >= 65 mmHg.
- **Implementation:** Draw **Lactic Acid** and **Blood Cultures.** Administer 30 mL/kg of crystalloid fluids for hypotension or lactate >= 4 mmol/L. Administer **Broad-Spectrum Antibiotics.**
- **Evaluation:** Lactate clearance, normalization of vital signs, improved urine output (> 0.5 mL/kg/hr).

Priority Topic 3: Child Health - Hematology/Oncology & Analgesics

Pathology Focus: Vaso-Occlusive Sickle Cell Crisis

The most common and painful complication of sickle cell anemia, caused by sickled erythrocytes obstructing microcirculation, leading to tissue ischemia, pain, and cellular infarction.

Diagnostic Labs & Monitoring

- **Hemoglobin (Hgb):** Baseline is typically low (7-10 g/dL); acute drops indicate aplastic or splenic sequestration crisis.
- **Reticulocyte Count:** Elevated due to rapid destruction of RBCs and compensatory bone marrow production.
- **Bilirubin:** Elevated due to RBC hemolysis.

🔵 Pharmacological Interventions

- **Opioid Analgesics:** 🔵 **Morphine** or 🔵 **Hydromorphone** via PCA pump. Avoid Meperidine due to the high risk of normeperidine-induced seizures.
- **Antineoplastic Agents:** 🔵 **Hydroxyurea** increases production of fetal hemoglobin (HbF), reducing the frequency of sickling crises.
- **Intravenous Fluids (IVF):** Aggressive hydration (e.g., 🔵 **0.45% Normal Saline**) to reduce blood viscosity and prevent further sickling.

🟦 ADPIE Case Study: The Pediatric Sickle Cell Patient

- **Assessment:** Severe, localized pain (often in extremities, back, or abdomen), swelling of joints (dactylitis in infants), fever.
- **Diagnosis:** Acute pain related to tissue ischemia; Deficient fluid volume.
- **Planning:** Achieve acceptable pain level (patient-defined), optimize hydration, prevent hypoxemia.
- **Implementation:** Administer high-flow IV fluids. Administer scheduled 🔵 **Morphine** around the clock (not PRN). Apply warm compresses to painful joints (do NOT use cold, as it promotes vasoconstriction and sickling). Provide supplemental oxygen *only* if hypoxic (SpO2 < 92%).
- **Evaluation:** Patient reports pain is manageable, oxygen saturation is maintained on room air, adequate and clear urine output.

🟪 Priority Topic 4: Adult Health - Cardiovascular System

🔴 Pathology Focus: Left-Sided vs. Right-Sided Heart Failure

Heart failure is a clinical syndrome characterized by the heart's inability to pump sufficient blood to meet the body's metabolic demands.

- **Left-Sided Heart Failure:** Primarily affects pulmonary circulation. The left ventricle fails to eject blood into the aorta, causing fluid to back up into the pulmonary veins and capillary beds, leading to pulmonary edema.
- **Right-Sided Heart Failure (Cor Pulmonale):** Primarily affects systemic circulation. The right ventricle fails to pump effectively, often as a result of left-sided failure or primary pulmonary hypertension, causing fluid to back up into the venous system.

🟢 Diagnostic Labs & Monitoring

- **B-Type Natriuretic Peptide (BNP):** > 100 pg/mL indicates heart failure. Elevated levels correlate directly with the severity of ventricular wall stretch and fluid overload.
- **Echocardiogram:** The gold standard diagnostic tool. Measures Ejection Fraction (EF). Normal EF is 55-70%. An EF < 40% indicates systolic heart failure.
- **Comprehensive Metabolic Panel (CMP):** Monitor 🟢 **Serum Potassium** strictly, especially when administering loop diuretics. Monitor 🟢 **Blood Urea Nitrogen (BUN)** and 🟢 **Serum Creatinine** for renal perfusion status.
- **Chest X-Ray (CXR):** Identifies cardiomegaly, pleural effusions, and pulmonary vascular congestion.

🔵 Pharmacological Interventions

- **Loop Diuretics:** 🔵 **Furosemide** or 🔵 **Bumetanide.** Administered to rapidly decrease preload by promoting renal excretion of sodium and water. *Crucial nursing action: Assess* 🟢 ***Serum Potassium*** *prior to administration due to the high risk of hypokalemia.*
- **ACE Inhibitors:** 🔵 **Lisinopril** or 🔵 **Enalapril**. Block the renin-angiotensin-aldosterone system (RAAS), promoting vasodilation and decreasing afterload. *Crucial nursing action: Monitor for the development of a dry, hacking cough or life-threatening angioedema.*
- **Beta-Adrenergic Blockers:** 🔵 **Metoprolol** or 🔵 **Carvedilol**. Decrease sympathetic nervous system stimulation, reducing heart rate and myocardial oxygen demand. *Crucial nursing action: Hold if heart rate is < 60 bpm or systolic blood pressure is < 100 mmHg.*
- **Cardiac Glycosides:** 🔵 **Digoxin**. Increases myocardial contractility (positive inotrope) and decreases heart rate (negative chronotrope). *Crucial nursing action: Assess apical pulse for one full minute before administration. Monitor for toxicity (visual disturbances, nausea, vomiting), which is exacerbated by hypokalemia.*

🟦 ADPIE Case Study: Acute Decompensated Heart Failure

- **Assessment:** Patient presents with severe dyspnea at rest, orthopnea, paroxysmal nocturnal dyspnea, crackles auscultated in bilateral lung bases, pink frothy sputum, and a respiratory rate of 28 breaths per minute.
- **Diagnosis:** Impaired gas exchange related to alveolar fluid accumulation; Decreased cardiac output related to altered contractility.
- **Planning:** Improve oxygenation status, rapidly reduce circulating volume (preload), and decrease myocardial workload.
- **Implementation:** Immediately elevate the head of the bed to high-Fowler's position. Administer supplemental oxygen to maintain SpO2 > 94%. Administer intravenous 🔵 **Furosemide** strictly over 1 to 2 minutes to prevent ototoxicity. Insert an indwelling urinary catheter to accurately measure output. Monitor telemetry continuously for ventricular dysrhythmias.
- **Evaluation:** Patient demonstrates unlabored respirations at 16 breaths per minute, clear lung sounds upon auscultation, oxygen saturation of 96% on room air, and a urinary output exceeding 100 mL/hr.

🟪 Priority Topic 5: Adult Health - Musculoskeletal System

🔴 Pathology Focus: Orthopedic Emergencies - Compartment Syndrome vs. Fat Embolism Syndrome (FES)

- **Compartment Syndrome:** A limb-threatening condition where increased pressure within a limited anatomical space (myofascial compartment) compromises circulation and tissue function. It leads to severe tissue ischemia and potential necrosis if untreated.
- **Fat Embolism Syndrome (FES):** A systemic and potentially fatal complication primarily associated with fractures of long bones (e.g., femur, tibia) or the pelvis, where fat globules enter the bloodstream and occlude microvasculature, most commonly in the lungs, brain, and skin.

🟢 Diagnostic Labs & Monitoring

- **Intracompartmental Pressure Monitoring:** Used to diagnose Compartment Syndrome. Pressures > 30 mmHg strongly suggest compromised capillary perfusion.
- **Arterial Blood Gas (ABG):** For FES, monitor for profound hypoxemia (PaO2 < 60 mmHg) and respiratory alkalosis progressing to acidosis.

- **Coagulation Studies:** 🟢 **Platelet Count** (often decreased in FES) and 🟢 **Prothrombin Time (PT) / Partial Thromboplastin Time (PTT).**
- **Creatine Kinase (CK) / Myoglobin:** Elevated in Compartment Syndrome due to severe muscle breakdown (rhabdomyolysis), which can lead to acute kidney injury.

🔵 Pharmacological Interventions

- **Analgesics (Opioids):** 🔵 **Morphine** or 🔵 **Fentanyl**. Required for severe pain. *Crucial nursing action for Compartment Syndrome: Note that the pain is typically unremitting and disproportionate to the injury, and is often unresponsive to these opioid analgesics. This is a primary clinical hallmark.*
- **Intravenous Fluids (IVF):** Aggressive crystalloid resuscitation (🔵 **0.9% Normal Saline**) is required in both conditions to maintain adequate mean arterial pressure and to flush myoglobin through the kidneys in the setting of rhabdomyolysis.
- **Corticosteroids:** 🔵 **Methylprednisolone.** Controversial but sometimes used prophylactically or as a treatment in FES to decrease the inflammatory response to fat globules.

🟦 ADPIE Case Study: The Post-Operative Femur Fracture

- **Assessment:** 24 hours post-open reduction internal fixation (ORIF) of a right femur fracture. The patient reports excruciating, burning pain in the right calf that is rated 10/10 and is not relieved by their scheduled dose of 🔵 **Hydromorphone.** The pain worsens dramatically with passive dorsiflexion of the right foot. Pulses are palpable, but capillary refill is sluggish (4 seconds).
- **Diagnosis:** Risk for peripheral neurovascular dysfunction related to increased tissue pressure.
- **Planning:** Prevent irreversible muscle and nerve ischemia. Facilitate immediate surgical decompression.
- **Implementation:** Immediately notify the orthopedic surgeon of the changing neurovascular status. Keep the affected extremity strictly at the level of the heart (do NOT elevate above the heart, as this decreases arterial perfusion pressure). Loosen any constrictive dressings, bivalve casts if present. Do NOT apply ice (causes vasoconstriction). Prepare the patient for an emergent fasciotomy.
- **Evaluation:** Following surgical fasciotomy, the patient reports a significant reduction in pain, capillary refill returns to < 2 seconds, and normal sensation is restored to the distal extremity.